AI OPPORTUNITY ANALYSIS

Recykal Marketplace

Where AI can realistically move the needle in a low-margin, high-volume scrap marketplace

Understanding the Business

WHAT THE COMPANY DOES, AND WHO PAYS

Recykal Marketplace is a B2B digital middleman between two sides of India's scrap trade: roughly 10,000 Waste Aggregators (small, single-owner operations that buy scrap from waste pickers and bale it) and about 1,000 Waste Recyclers (large industrial plants that buy baled scrap and process it into raw material). Recykal buys from aggregators and sells to recyclers, so both sides effectively pay for what the platform provides: aggregators accept a slightly lower price in exchange for fast, guaranteed payment; recyclers pay for GST-compliant, quality-checked supply they can source in bulk without chasing hundreds of informal sellers themselves.

HOW IT MAKES MONEY

Recykal takes ownership of the trade rather than just facilitating it. It buys baled scrap from aggregators and resells it to recyclers, capturing the spread between the two prices. It is a low-margin, high-volume business by nature: in FY23-24 the marketplace did roughly INR 750 Cr of business at about 1.5% net margin, so pricing, quality, and collections all have to be tight for the business to stay profitable.

WHAT IT HAS TO GET RIGHT

- Matching supply to demand quickly and at the right price, since the purchase team decides which seller's lot goes to which buyer.
- Paying sellers fast (3-7 days) while buyers often pay late, carrying the working-capital gap between the two.
- Enforcing and proving quality with geo-tagged photos, Recykal's main leverage when negotiating debit notes.
- Enforcing GST compliance on the seller side through tax-portal integration, because buyers specifically value this.
- Arranging reliable trucking directly from the seller's location.
- Keeping a detailed transaction log, since disputes on both sides get settled by referring back to it.

WHERE IT BREAKS DOWN

- Buyers delay payments despite agreed terms, forcing repeated manual collections follow-ups and occasional supply holds.
- Debit notes: quality checks happen at the buyer's plant with no seller present, so Recykal must negotiate after the fact how much it absorbs versus passes to the seller.
- The cash-flow gap (paying sellers in days, collecting from buyers in weeks) creates working-capital strain that scales with volume.
- With only ~1.5% net margin, small errors in quality assessment, pricing, or collections timing have an outsized effect on profitability.
- Prices are volatile and outside Recykal's control: PET pricing tracks petroleum prices, and moisture (treated as an impurity) rises in monsoon and winter, pulling down both price and transaction volume.
- Aggregators are informal SMEs, so GST non-compliance and inconsistent documentation add friction and risk on the sourcing side.

PRODUCT LESSON *Exposing internal predictions to the market can backfire. The price prediction model was originally shown to sellers in-app to counter PETScrapwala's price-discovery dominance, but once aggregators leaned on it, they began quoting Recykal's own predicted prices back at Recykal during negotiations, so it was withdrawn from seller visibility and kept for internal buy-sell decisions only.*

DATA COLLECTED ALONG THE WAY

- Transaction data: quantities, agreed prices, buyer and seller identities, dates.
- Geo-tagged photos of material quality captured at both pickup and delivery.
- GST compliance status per seller, pulled via tax-portal API integration.
- Payment timelines and delay patterns per buyer.
- Debit note and quality-dispute history.
- Logistics data: truck bookings and delivery timing.
- A detailed log tying all of the above together per transaction.

Proposed AI Features

Feature	AI Type	Problem It Fixes / What It Does	Data Needed	Metrics Moved
Auto Quality Grading	Image AI (Computer Vision)	Scores bale purity/contamination from the geo-tagged pickup photos already being captured, before the load reaches the buyer.	Historical geo-tagged photos, matched contamination/purity labels, and past debit note outcomes tied to those photos.	% of debit notes disputed, average debit note value, dispute resolution time.
Buyer Payment Risk Score	Prediction AI (ML)	Flags recyclers likely to delay payment before the sale, so collections and supply decisions are proactive, not reactive.	Buyer payment history (on-time vs delayed days), invoice sizes, order frequency, market price trends.	Days Sales Outstanding (DSO), collections follow-up volume, working capital tied up.
Smart Seller-Buyer Matching	Recommendation AI (Next Best Buyer)	For each seller's lot, recommends the single best-fit buyer out of many, ranked on price, quality grade, and payment reliability.	Live inventory by material/region, buyer demand signals, historical outcomes, quality grades, buyer risk score.	Net margin per transaction, time-to-match, fill rate.
Debit Note Settlement Classifier	Prediction AI (ML) - Classification	Recommends a settlement action for each incoming debit note (accept in full / partial accept at X% / contest) to Recykal's negotiation team, using buyer-seller relationship history, the Auto Quality Grading score from the same shipment, the debit note's claimed reason and amount, and the buyer's payment mode (deferred-payment buyers have more room to use debit notes as an unofficial discount lever). Runs internally only, never shown to buyers, the same lesson learned with price prediction.	Buyer/seller relationship history (transaction volume, past debit note frequency and outcomes), quality-grading scores per shipment, debit note text/amount/reason, buyer payment mode and terms, historical settlement outcomes for training.	% of debit note value recovered, negotiation cycle time, margin protected, model accuracy vs. actual negotiated outcome.

Feature	AI Type	Problem It Fixes / What It Does	Data Needed	Metrics Moved
PET/Scrap Price Prediction	*Prediction AI (ML)*	*Forecasts near-term prices and volumes from petroleum-price correlation and seasonal moisture effects, so purchasing can be timed to protect margin. Already live at Recykal; originally shown to sellers to counter PETScrapwala, later withdrawn to internal-only use once sellers began quoting the predictions back in negotiations.*	*Historical PET/scrap prices, crude/petroleum feeds, moisture readings per lot, rainfall/humidity by region, seasonal transaction volumes.*	*Net margin per transaction, forecast accuracy, volume planned around seasonal dips.*

Recommended Build Priority

I would build the Auto Quality Grading feature first. Recykal already captures the exact data this needs (geo-tagged photos at pickup and delivery), so it doesn't require new data collection, just a labeling effort on existing history. Debit note disputes sit directly on the business's thinnest point, a 1.5% net margin, so even a modest reduction in disputed value or faster resolution has an outsized effect on profitability. It also strengthens the two things Recykal actually sells: fast payment to sellers and reliable quality to buyers. In practice this could be phased in as a supporting score to the purchase team's decisions rather than a full replacement, which keeps the risk low while it proves out.

The payment risk model would be my second choice: it targets the other core pain point, the cash-flow gap, but depends more heavily on having enough clean payment history per buyer to be reliable. Notably, the PET/Scrap Price Prediction model is not hypothetical, it was actually built and used to plan transaction timing around petroleum price movements and seasonal moisture effects, and it's the clearest proof point that AI already moves the margin needle in this business. It also carries a useful lesson for the rest of this roadmap: any feature that surfaces Recykal's internal judgement to a counterparty in a negotiation, including the Buyer Payment Risk Score or Smart Seller-Buyer Matching, should be checked against this same exposure risk before launch. The Debit Note Settlement Classifier is built internal-only from the start for exactly this reason.